

PCB Camera Inspection Acceptance Specification

Example requirements for detector findings, deterministic disposition, and human review

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Owner	Quality Engineering	Effective	13 September 2026
Applies to	PCB optical inspection cell	Status	Example / not production

1. Purpose and decision authority

This example defines image-level acceptance logic for a PCB optical inspection cell. The detector supplies candidate regions and confidence scores. The approved deterministic policy supplies the final PASS, FAIL, or HUMAN REVIEW disposition.

1.1. Approval gate

Rules extracted from this document are draft candidates only. A quality engineer must verify the cited source evidence and explicitly approve the complete policy before it can control an inspection.

2. Defect-class requirements

2.1. Critical circuit defects

Any open-circuit (OP) or short-circuit (SH) finding is critical and must result in FAIL.

2.2. Conductor scratches

A conductor-scratch (CS) finding requires HUMAN REVIEW before disposition.

2.3. Cosmetic spur allowance

A spur (SP) finding is acceptable for this demonstration and may PASS when no higher-priority rule applies.

2.4. Maximum finding count

More than two actionable PCB defect findings in one image must result in FAIL.

This document is a portfolio example, not a factory acceptance standard. A client must replace these limits with validated production requirements.

Confidence and capability controls

Uncertain or unsupported requirements fail safely to human review

3. Detector confidence

3.1. Unconfirmed findings

Findings below 30 percent confidence may be ignored. Findings from 30 percent up to but below 70 percent confidence require HUMAN REVIEW.

4. Scope boundary

Calibrated length measurements and missing-component checks are outside this example policy. A client deployment must add camera calibration, a reference assembly, and validated detector capabilities before those requirements can be automated. Any future requirement that the active detector or rule engine cannot evaluate must produce HUMAN REVIEW rather than an automatic PASS or FAIL.

Disposition	Meaning	Required action
PASS	No fail or review rule triggered	Release under approved process
FAIL	One or more fail rules triggered	Reject and record decisive evidence
HUMAN REVIEW	Uncertain or unsupported requirement	Quality engineer resolves disposition

Approval record

Quality engineer	_____
Approval date	_____
Approved policy version	_____